

Agrya Halder

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Work Experience

Indian Statistical Institute (ISI) Kolkata

Kolkata, India

Project Linked Personnel (PLP)

Mar 2024 - Jul 2025

[advised by **Prof. Sanghamitra Bandyopadhyay** and **Dr. Abhik Ghosh**]

- Recovery of cell locations in Spatial Transcriptomic (ST) sequences.
- Effects of outliers in predicting domain layers of single cell RNA sequences.
- Robust generation of ST data in order to improve model prediction.

Indian Institute of Technology(BHU)

Varanasi, India

Senior Research Fellow (SRF)

Sep 2023 - Jan 2024

- Leading a team of 2 postgraduate students in order to solve MOT challenge and planning challenges like Multi-person Tracking (accepted @ Multimedia Tools and Appl.), occlusion reconstruction (accepted @TNNLS IEEE), and Suspicious Activity Recognition.
- Extended the Gait de-identification with the inclusion of pose substitution network in order to achieve 30% (SIVP 2022) less recognition accuracy.

Indian Institute of Technology(BHU)

Varanasi, India

Junior Research Fellow (JRF)

Sep 2021 - Aug 2023

- Collaborated with Prof. Shatish Kumar (Washkewicz College of Engineering, Ohio, USA) to develop a gait de-identification model.
- Significantly boosted the de-identification score in terms of reducing the recognition accuracy of de-identified frames.

Sciffer Analytics Pvt. Ltd.

Pune, India

Data Science - Intern

Nov 2020 - Feb 2021

- Collaborated with a five-person team to develop a CNN model that utilised FRCNN as a foundation to improve the accuracy of real life data classification.
- Collaborated with data collection team in order to collect a significant amount of dataset for image detection purpose.
- Involved in a team to optimize operational research problems.

Education

Indian Institute of Technology, Madras

Jul 2025 - Current

PhD in IIT Madras - Deakin University Research Academy

- **Topic:** Assessment of mental fatigue using multi-modal feature fusion

Banaras Hindu University

August 2018 - July 2020

Master in Science (MSc) in Computer Science

8.09/10.0

- **Graduated with First Class** (top 2% on overall achievements out of the class)
- Thesis: Multimodal Sentiment Analysis using Multi-head Cross-modal Attention

University of Calcutta

July 2015 - June 2018

Bachelor in Science (BSc) Honours in Computer Science

First Class

- Passed with First Class
- Specialised in Physics and Maths with Computer Science as major

Selected Projects

Future Frame Prediction

Varanasi, India

SRF @ IIT(BHU)

2022 - 2023

- Generative Spatial Self-Attention Transformer Unit (GSSTU), a novel approach for future frame prediction has been proposed by combines the Transformer's adaptation strength with the 3D Convolutions.
- Further introduced a self-attention-based spatial feature preservation mechanism that further follows the novel neighborhood pixel intensity loss to preserve the structural quality, thereby enhancing the quality of the predicted frames.
- our model exhibited an average improvement of 4.64%, 18.5%, and 42% concerning SSIM, PSNR, and LPIPS metrics correspondingly across five benchmark datasets.

Gait De-identification

Varanasi, India

JRF @ IIT(BHU)

2021 - 2022

- Proposed GtNet, a plausible deep learning-based solution to the gait de-identification problem.
- Our model obfuscates the identifiable gait features (i.e., walking pattern of an individual) by simultaneously retaining the pose and color/texture information.
- GtNet takes as input an RGB frame along with its dense pose features and the dense pose features of a target frame with a similar walking pose and outputs a translated RGB frame with a different structural appearance.
- Our model achieves a significant gait recognition accuracy drop of upto 24% on SOTA gait recognition models

Multimodal Sentiment Analysis using Multi-head Cross-modal Attention

Varanasi, India

Master Thesis - Major

2019 - 2020

- Analysing Multimodal data (combination of Video, Audio, and Texts) from CMU-MOSEI dataset and discover patterns in the data that show trends between positive and negative sentiments.
- Re-design the network with the help of self-attention layers of transformer blocks to outperform the state-of-the-art accuracy score in terms of classification.
- Our model claimed to increase the accuracy by 2% in aggregated feature representation.

Skills

Programming Python (Pandas, NumPy, Scikit-learn. etc.), C/C++, HTML/CSS, MatLab, SQL.

Framework PyTorch, TensorFlow, OpenMP

Miscellaneous and Tools Linux, Shell (Bash/Zsh), $\text{\LaTeX}$ (Overleaf), Microsoft Office, Git, draw.io, GIMP

Achievements

2023 **Best Student Paper**, Presented and awarded as Best Paper on CVIP'2023 at IIT Jammu.

India

2023 **DST**, Awarded a travel grant to present paper at CVIP'2023

India

2022 **SERB DST**, Awarded a travel grant to attend ICVGIP'2022

India

2021 **JRF**, joined as junior researcher at IIT(BHU)

India

2020 **GATE**, Qualified

India

2019 **Silver**, Applied Natural Language Processing NPTEL Exam

India

Peer-reviewed Publications and Patents

JOURNAL ARTICLES

GSSTU: Generative Spatial Self-Attention Transformer Unit for Enhanced Video Prediction [[Paper](#)]

Binit Singh, Divij Singh, Rohan Kaushal, **Agrya Halder***, Pratik Chattopadhyay

IEEE Transactions on Neural Networks and Learning Systems 2024

A Systematic Survey on Recent Deep Learning based Approaches to Multi-Object Tracking [[Paper](#)]

Harshit Agarwal, **Agrya Halder***, Pratik Chattopadhyay

Multimedia Tools and Applications 2023

STAFU: Spatio-Temporal Attentive Fusion Unit for Future Frame Prediction in Video Sequences [[First revision](#)]

Binit Singh, **Agrya Halder***, Pratik Chattopadhyay

Expert Systems With Applications 2023

BGaitR-Net: An Effective Neural Model for Occlusion Reconstruction in Gait Sequences by Exploiting the Key Pose Information [[Paper](#)]

S.S. Kumar, Binit Singh, Pratik Chattopadhyay, **Agrya Halder***, Lipo Wang

Expert Systems With Applications 2023

GTNet: Gait Transformation Network for Gait De-identification with Pose Preservation [[Paper](#)]

Agrya Halder*, Pratik Chattopadhyay, Sathish Kumar

Signal, Image and Video Processing (SIVP) 2022

CONFERENCE PAPERS

MotionFormer: An Improved Transformer-Based Architecture for Multi-Object Tracking [[Best Student Paper](#)]

Harshit Agarwal, **Agrya Halder***, Pratik Chattopadhyay

The 8th International Conference on Computer Vision & Image Processing (CVIP-2023)

References available upon request.